Binding of VEEV inhibitors to Impα nuclear import protein using molecular docking

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Introduction

Importing host proteins into nucleus is performed by Impβ/Impα complex, which recognizes and binds NLS sequence regions in proteins. Venezuelan equine encephalitis virus (VEEV) is a neurotropic alphavirus which is a biodefense concern because of its potential for aerosol spread and lack of sufficient remedies. VEEV capsid protein contains NLS-like fragment, which competes with host proteins for the NLS binding site on Impα protein. A possible solution for blocking VEEV infection cycle could be designing a molecular inhibitor, which binds to the NLS binding site of Impα and prevents interactions with VEEV protein.

Methods

G281-1485 is a lead candidate inhibitor to block capsid binding to Impα. We examined its binding and of its variants to the Impα using molecular docking.

1. Using CASTp, we identified nine potential binding sites on Impα protein (PDB ID 3VE6) in addition to the NLS binding site.
2. Using AutoDock Vina docking simulations, we studied the binding affinities of VEEV inhibitors.

Results

Using multiple AutoDock simulations we found the following:

- Inhibitor G281-1485 (WT) binds to Impα at multiple locations shown in Fig. 1 and Table 1. NLS binding site has the highest affinity for the WT supporting its inhibitory activity by interfering with binding of VEEV NLS sequence. Interestingly, all ligand modifications bind with highest affinity to NLS binding site.
- We considered seven modifications of G281-1485 shown in Fig. 2, which change its hydrophobicity or add hydrogen donors or acceptors. We found that all hydrophobic modifications reduced the binding free energy G (Table 2). The best is mod2+3 extending the ligand arm with hydrocarbon and replacing oxygen with CH2 group. Eliminating one hydrocarbon from the ligand arm (G281-1481) compromises it binding affinity increasing G. Replacement of oxygen with carboxyl group CO reduces G (O2CO). Adding amino group to the ligand arm terminus does not improve binding (C17NH2).

Results (continued)

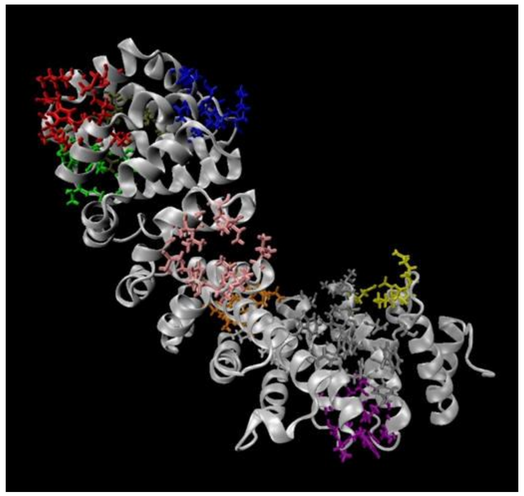

Figure 1: NLS binding site and nine potential binding sites on Impα.

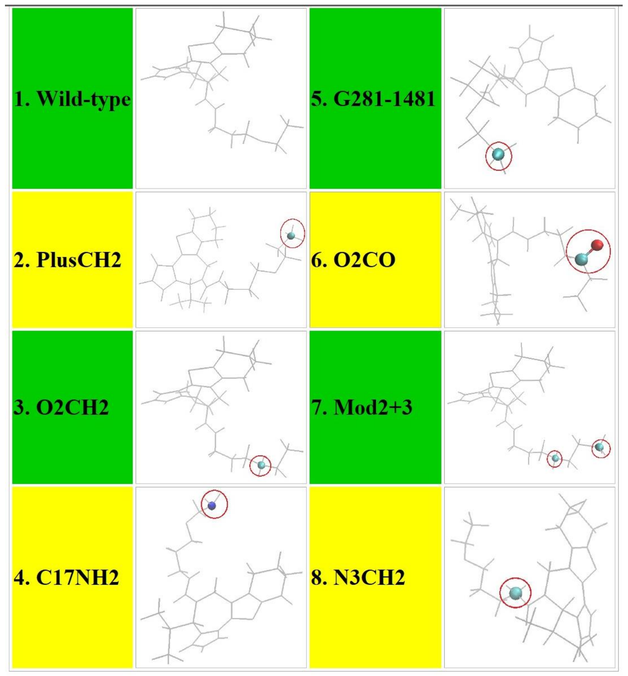

Figure 2: WT and modified inhibitors studied by docking.

Table 1: Binding affinities of WT inhibitor vs. 10 binding sites on Impα

	Wild-type
Binding Sites	32-run Best Affinity (kcal/mol)
NLS	-6.5
1	-5.3
2	-5.3
3	-5.1
4	-4.7
5	-5.1
6	-4.4
7	-4.7
8	-5.2
9	-5.1

Table 2: Binding affinities of WT and its modified ligands vs. NLS BS

	NLS BS
Inhibitors	32-run Best Affinity (kcal/mol)
Wild-type	-6.5
PlusCH2	-6.6
C17NH2	-6.3
O2CH2	-6.7
G281-1481	-6.2
O2CO	-6.9
Mod2+3	-7.0
N3CH2	-6.6

Conclusions

1. Although there are multiple binding sites for G281-1485 inhibitor on Impα surface, the WT inhibitor binds with highest affinity to NLS binding site.
2. All modifications enhancing hydrophobicity improve binding affinity. Polar modifications have the opposite effect.
3. These results will guide the search for better inhibitors of VEEV capsid binding to Impα.